

Total No. of Questions : 6]

SEAT No. :

P149

[Total No. of Pages : 2

APR. - 16/BE/Insem. - 69

B.E. (Information Technology)

MOBILE COMPUTING (414463 A)

(2012 Course) (Semester - II) (Elective - III)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Draw and explain basic PCS architecture. **[4]**

b) Write a Short Note on: **[6]**

- i) VLR Database Overflow.
- ii) Cell Splitting.

OR

Q2) a) What are the differences between 1G, 2G and 3G mobile technologies? **[5]**

b) What is Handoff and what is roaming? Draw the message flow executed during adding new BS with soft handoff. **[5]**

Q3) a) Explain the three algorithms used for providing security in GSM i.e. A3, A5 & A8. How is subscriber authentication and data Encryption done in GSM. **[6]**

b) State the main functions of : **[4]**

- i) BTS and BSC,
- ii) MSC / GMSC,
- iii) OMC, and
- iv) NMC

OR

P.T.O.

- Q4)** a) Describe logical and physical channels and frame hierarchy of GSM with neat sketch. [5]
- b) List and describe the various issues in location update procedure in GSM, provide solution to handle any one issue in detail. [5]

- Q5)** a) Describe the following types of short messages. [6]
- i) User Specific.
 - ii) ME Specific.
 - iii) SIM Specific.
- b) What are various strengths of SMS? Explain all of them? Also state what are the applications areas where these strengths can be used? [4]

OR

- Q6)** a) Explain mobile terminated messaging and origination in SMS architecture. [5]
- b) Draw and explain fixed network number portability. [5]

